Vigneshwara Chinnadurai

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Professional Summary

Senior AI Architect & Data Scientist with **8+ years** of experience designing enterprise-scale **Agentic AI, Generative AI**, and Machine Learning solutions. Architects production-grade **multi-agent systems** on **Google ADK** and **AWS AgentCore SDK**, with expertise in **Hybrid RAG pipelines**, **vector databases**, and **LLM self-reflection (self-critique) loops**. Delivered **$277M+ measurable business impact** across Fortune 500 clients.

Experience

AI/ML Computational Science Associate Manager — June 2026 – Present
Accenture Solutions — *Bangalore, India*

Aon Human Capital – Claims Audit Modernization (Agentic Platform) | *Google ADK, Azure OpenAI, LiteLLM, Azure Container Apps, PostgreSQL/Databricks, Zero-Trust JWT*

- Co-authored the **end-to-end enterprise solution architecture blueprint** (C4 Context/Container/Component views + Azure hub-and-spoke deployment) for a **24-agent multi-agent platform on Google ADK** (1 Orchestration + 12 Super + 11 Utility agents) automating the full claims-audit lifecycle – **100% automated claim-vs-plan-provision comparison**, configurable stratified sampling, and discrepancy reporting – replacing a manual, spreadsheet-driven audit process.
- Architected a **deterministic three-tier hierarchical delegation model** (Orchestrator → Super Agents → Utility Agents) on **Google ADK** with zero LLM routing discretion, routing all model traffic through a **LiteLLM → Aon GenAI Wrapper → Azure OpenAI (gpt-5.1 / gpt-5-mini)** path so no agent ever calls the model provider directly – defended the framework's cloud-agnostic design (via the LiteLLM abstraction) to the review board as future-proofing against a potential GCP shift.
- Designed the **shared Postgres blackboard data plane** (agents exchange row/file references, never bulk data) with a planned store-swap to **Databricks (Delta / Unity Catalog)**, and a **three-plane runtime model** (Control / Data / Memory) that isolates containerized inter-agent invocation from persisted state, enabling horizontal scaling with zero state migration.
- Defined the **zero-trust JWT security model** – 5-minute TTL, per-call Okta-issued tokens validated at APIM, RBAC propagated across the full agent call tree – and **mechanically-enforced Human-in-the-Loop gates** (Supervisory Control, Approval, Findings Review, Exception Escalation) that hard-stop the pipeline via a `PAUSED` sentinel independent of LLM behavior.
- Established the **MCP (Model Context Protocol) integration pattern** behind APIM for 6 line-of-business systems (SharePoint, MOVEit, Outlook, Rules Engine, Claims Datastore, DocuSign) eliminating point-to-point integrations, plus the **Agent Catalog governance model** (Agent Signature metadata schema, semantic versioning, lifecycle states, automated stewardship for idle/orphaned agents) across Aon's Human Capital Azure subscription.

Senior AI Architect (AI and Data Science Engineer III) — January 2022 – June 2026
Deloitte Consulting USI — *Bangalore, India*

Disney TVA Schema Intelligence Chatbot | *AWS Bedrock, Claude 3.7 Sonnet, Agent Forge SDK*

- Architected a **production-grade RAG chatbot** using **custom S3VectorsAgentRuntime** extending AWS AgentCore SDK to handle S3 Vectors' unique `metadata.text` storage pattern that standard SDK couldn't parse.
- Implemented **intelligent token management** with configurable `max_rag_context_size` trimming that dynamically truncates retrieval chunks to prevent context overflow.
- Built **dual-path KB retrieval**: SDK's `extract_retrieval_chunks` for standard `content.text` + custom fallback for `metadata.text` with automatic deduplication.
- Created **follow-up query resolution system** with heuristic-based pronoun detection + URN extraction, rewriting vague queries for improved RAG retrieval.
- Optimized memory retrieval latency from **17+ seconds to under 5 seconds** through memory hook optimization and session stashing before RAG augmentation.
- Built **KB write-back pipeline** with deterministic vector keys enabling immediate queryability of agent-generated summaries without sync delays.

Disney PGT Spec Sheet Planner Agent | *AWS Bedrock AgentCore, Strands SDK, Claude Sonnet, pgvector*

- Architected **11-tool deterministic AI pipeline** (1,550+ LOC) on **AWS Bedrock AgentCore** for automated spec-sheet field extraction—parsing Excel/PDF/DOCX/image files, extracting structured network data (budgets, flight dates, unit lengths) via Claude Sonnet, and writing to PostgreSQL with **Human-in-the-Loop** approval via Teams Adaptive Cards.
- Achieved **5.3x end-to-end latency reduction** (237s to 44.9s) by eliminating cold starts, optimizing content chunking, and reducing extraction time from **39s to 9.2s** (4.3x faster).

- Eliminated **33 seconds of LLM reasoning overhead** (74% of E2E) by replacing Strands agent tool-routing with a **deterministic sequential pipeline**—zero quality degradation, pure latency savings.
- Engineered **aggressive parallel execution** using `ThreadPoolExecutor`: parse || RFP-lookup, intermediate-write || extraction, save-final || DB-write—with **adaptive Bedrock retries** and chunked extraction fan-out across 4 parallel workers.
- Built **pgvector (vector DB) integration** with Azure OpenAI `text-embedding-3-large`, **Claude Vision OCR fallback** for image-based spec sheets, and **RFP context resolution** with DB-driven network filtering reducing LLM input by up to 60%.

Disney ESPN Game Summary Platform | *AWS Nova Premier, Lambda, Real-time Video AI*

- Developed **GenAI highlight extraction platform** contributing to **$231.76M AWS business expansion**. Architected **7-step serverless Lambda pipeline**: MediaLive → Kinesis → MediaPackage → S3 → Nova Premier → DynamoDB → Highlights.
- Designed **multimodal prompt engineering** for video analysis with confidence scoring per field, play-type classification, and precise timestamp extraction. Built `clean_response_text()` handling markdown fences and partial JSON.
- Validated **approx. 90% event capture rate** on the **NFL (American Football)** MVP; proposed expansion to **Cricket, Rugby, and Soccer** pending client validation of results.

HP Enterprise - CXO Talk to Data Platform | *Databricks Genie, Azure OpenAI GPT-4o-mini*

- Built **AI-powered conversational analytics platform** replacing traditional BI dashboards for executive decision-making. Implemented **Databricks Genie NL-to-SQL** with user token forwarding.
- Architected **multi-table semantic layer** (2 Fact + 11 Dimension tables) with YAML-based semantic definitions. Implemented **hallucination reduction** through output validation against schema definitions.

Toyota Demand Forecasting Engine | *LSTM, N-BEATS, Prophet, PySpark*

- Engineered enterprise-scale forecasting system managing **1.2 million SKU combinations** across **2 Part Centers (PCs) and 15 Part Distribution Centers (PDCs)**, reducing annual inventory costs by **$46.1 Million** at **70% forecast accuracy** (1 – WMAPE).
- Implemented **hybrid LSTM architecture** with dual-layer Lambda expansions, **zero-entry "+1 trick"** for MinMax scaling, and **ensemble model selection** (LSTM/N-BEATS/SVR/Prophet) per SKU characteristics.
- Designed `orchestrator.py` with **multiprocessing pool** executing parallel forecasts with YAML-driven multi-environment configuration.

Halliburton Financial Predictive Modeling | *N-BEATS, SARIMAX, Deep Learning*

- Developed **deep learning forecasting solution** for Accounts Payable/Receivable prediction using **N-BEATS** and **SARIMAX** ensemble, achieving **98% accuracy** in invoice dispute prediction.
- Reduced financial forecast **delivery time by 98%** through automated pipeline optimization and multi-variate time series modeling capturing both trend and seasonality patterns.

Firm Initiatives: Agentic AI & MLOps Estimation Frameworks

- Created **estimation frameworks** for Agentic AI and MLOps project sizing, refined over 24+ iterations and **adopted across multiple Deloitte practices** for consistent project scoping.

Senior Software Engineer (Machine Learning) February 2018 – January 2022
WIPRO Technologies *Bangalore, India*

WIPRO Holmes (R&D) | *LSTM, Reg-exp, Deep Learning*

- Enhanced Contract Review tool using **Bi-LSTM + CRF architecture** for sequence labeling of legal clauses in MSAs and SOWs. The **bidirectional LSTM** captured long-range contextual dependencies while the **CRF layer** learned valid BIO tag transitions (B-CLAUSE → I-CLAUSE), preventing invalid sequence predictions. Combined with **character-level embeddings** for OOV legal terminology and **domain-specific regex patterns** for structured entities (dates, parties, amounts), improved extraction accuracy from **76% to 93%**.

Huawei Video (R&D) | *DIP, Vector DB, Deep Learning*

- Created data model for **real-time recommendation pipelines** using **Deep Interest Network (DIN)** for user behavior modeling. Built **attention-weighted embeddings** capturing user-item interactions, stored in **Redis** for low-latency serving + **Milvus** for vector similarity retrieval. Integrated with **RASE (Huawei Recommendation-as-a-Service)** for 3 pipelines: **Hot Broadcast** (trending long-form), **You May Like** (personalized long-form), and **Waterfall E2** (short-video infinite scroll).

Shell Plc | *CNN, Transfer Learning, Machine Learning*

- Built **CNN with VGG19 Transfer Learning** in Keras to detect small symbols (valves, pumps, instruments) in P&ID drawings. Implemented **Hough Circle Transform** + **TensorFlow Object Detection API** for geometric shape recognition, automating 2D-to-3D schematic conversion.
- Developed **OCR pipeline** combining **Tesseract** (for structured text) + **AWS Textract** (for tables/forms) with custom image preprocessing (contrast enhancement, adaptive thresholding, noise reduction) for legacy engineering document digitization.

Personal R&D Projects

Autonomous Multi-Agent Strategic Analysis Engine | *Gemini 2.5 Flash, Hybrid RAG, Self-Refinement*

- **LLM-driven Task Planning**: `AnalystAgent` decomposes complex prompts into executable multi-step plans at runtime, with no hardcoded workflows.
- **Hybrid Retrieval Pipeline**: Custom BM25 sparse + Gemini dense embeddings with **Reciprocal Rank Fusion (RRF)** + **LLM-powered reranking** (0-1 relevance scoring).
- **Self-Reflection & Refinement Loop**: Rubric-based self-critique (Depth/Evidence/Causality/Actionability) with automatic `NEEDS_REFINEMENT` detection triggering re-retrieval cycles.
- **Dynamic Tool Generation**: Runtime code synthesis creating callable Python functions from LLM output. **Graph Memory** with semantic triple extraction for persistent knowledge.
- **Real data integration**: SEC EDGAR XBRL, Google News RSS, Gemini Search Grounding. **Cost Guard System** with budget caps and aggressive caching.

Quantum Portfolio Optimization Engine (MBA Project) | *D-Wave Quantum Annealing, QUBO, Simulated Annealing*

- **QUBO Formulation from First Principles**: Quadratic Unconstrained Binary Optimization with risk terms (covariance), return rewards (mean returns), and cardinality constraints via penalty functions.
- **Classical-Quantum Benchmark**: Side-by-side scipy max-Sharpe vs. quantum annealing with hyperparameter sweep over *risk & return*, penalty coefficients.
- **D-Wave Leap QPU Integration**: Production-ready execution with `EmbeddingComposite` for automatic qubit mapping. Efficient frontier visualization.

LeetCode AI Companion (Android) (Personal Development) | *Kotlin, Jetpack Compose, Gemini API, GraphQL*

- **Jetpack Compose UI** with Material Design 3. LeetCode GraphQL integration for daily challenges. **Gemini generation pipeline** with retry logic.
- **GitHub Push Automation**: One-tap export of solutions to repository. Calendar event integration for completion tracking. Mermaid flow viewer with gestures.

Education

Master of Business Administration, Analytics and Data Science | Manipal University, Jaipur, India | August 2024 – June 2026
Bachelor of Engineering, Electrical & Electronics | Anna University, Chennai, India | August 2014 – May 2017

Certifications

AI Fluency: Framework & Foundations | Anthropic (Mar 2026)

Claude 101 | Anthropic (Mar 2026)

AWS Certified Machine Learning – Specialty | Amazon Web Services (Oct 2021)

Delivery Accreditation - AI Agents | ServiceNow

Configure AI Applications to Optimize Search Results | Google

Deep Learning Specialization | Coursera/Udacity

Awards & Recognition

The Doer Award | Deloitte Consulting (2026)—recognized for turning an uncertain idea into a production reality

Technical Skills

Agentic AI & RAG: Google ADK, Agent Forge SDK (AWS AgentCore), Strands Agents SDK, Multi-Agent Systems & Orchestration, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25 + Dense + RRF), LLM Reranking, Vector Databases (S3 Vectors, pgvector, Milvus), Token-aware Trimming, LLM Planning & Critique, Self-Reflection Loops, Hierarchical Agent Delegation, Deterministic Pipeline Orchestration, LiteLLM, Model Context Protocol (MCP), Zero-Trust JWT, Human-in-the-Loop Gate Design, Agent Catalog & Lifecycle Governance

LLMs & Foundation Models: Claude 3.7 Sonnet, GPT-4/4o-mini, AWS Nova Premier, Gemini 2.5 Flash, Llama 3, Amazon Titan Embed v2

Quantum Computing: D-Wave Quantum Annealing, QUBO Formulation, Simulated Annealing, D-Wave Leap, Binary Quadratic Models

Deep Learning: Transformers (BERT, DiT), CNN (VGG19, ResNet), LSTM, Bi-LSTM, Deep Interest Network (DIN), N-BEATS, Prophet

Cloud & MLOps: AWS (Bedrock, SageMaker, Lambda, S3, MediaLive, Kinesis, SQS, Identity Manager, Media Package, Media Convert), Azure (OpenAI, Container Apps), Google Cloud / GCP (Vertex AI), Docker, Kubernetes, Airflow, DataBricks, Okta

Languages & Data: Python, PySpark, SQL, Kotlin (Android), Pandas, NumPy, TensorFlow, PyTorch

Key Achievements

$46.1M annual inventory savings at Toyota | **$231.76M** AWS expansion via Disney ESPN | **5.3x** pipeline latency reduction (237s to 44.9s) | **74%** overhead eliminated via deterministic orchestration | **93%** contract extraction accuracy | **1.2M SKUs** forecasting scale | **4+ Fortune 500** production deployments